Learned Deferral for Auditable Bounded Decision Control

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Abstract

Production AI systems often operate with incomplete, conflicting, or insufficient evidence. Forced classifiers collapse such cases into action labels, while generative systems can produce outputs that are difficult to interpret as auditable execution decisions. This paper presents EvaluatorDPT, a bounded decision-control model that predicts YES, NO, or TBD, where TBD is learned as a deferral outcome rather than added only as a post-hoc confidence rule. The model uses a transformer encoder with a primary bounded-decision head and structured auxiliary channels for values and emotions/sentiments. For the current S12B publish candidate, we report decision performance on DS-L validation and test splits; auxiliary emotion metrics are omitted because the emotion head is masked in this lineage. On the full DS-L test split (n=44,597), S12B achieves Accuracy = 0.8260 and Macro F1 = 0.8252, with per-class F1 of 0.8314 (YES), 0.8486 (NO), and 0.7956 (TBD). The release also includes calibration evidence (ECE = 0.0338 on validation), threshold-sweep artifacts, multi-seed stability checks, confusion matrices, reproducibility commands, and a certifier-facing artifact pack. The main contribution is a bounded decision interface in which deferral is learned, operating thresholds remain inspectable, and release artifacts are organized for external review.

1 Introduction

Many deployed systems now connect language inputs to downstream actions. In such settings, classification accuracy alone is not the central question; the system must also decide whether to act, block, or defer. Inputs may be incomplete, contradictory, or ambiguous. A model that must always choose YES or NO can conceal uncertainty in cases where an operator would prefer review.

The issue appears when language models or automated agents are connected to business workflows, policy enforcement, moderation, approvals, safety checks, or other action-taking systems. In these environments, a model output is not merely a label. It becomes a control signal. A false YES may trigger an unsafe or unauthorized action; a false NO may block a valid action; and a failure to defer may hide uncertainty from the surrounding workflow. A useful decision model therefore needs an explicit representation of uncertainty, rather than a confidence score attached to a forced label.

EvaluatorDPT is designed for this setting. It is not a general-purpose language generator or a replacement for domain-specific review. It is a bounded routing layer that converts a context signal into a constrained decision with confidence and supporting artifacts. Downstream policies can inspect the decision distribution, choose operating thresholds, route deferred cases, and preserve a review trail.

EvaluatorDPT addresses this problem by modeling decision control as a bounded three-way output space:

$$\mathcal{Y}_d = \{\mathrm{YES}, \mathrm{NO}, \mathrm{TBD}\}.$$

The TBD outcome is not merely a post-hoc rejection threshold. It is a learned class that allows the model to internalize deferral boundaries during training. Thresholding remains useful at deployment time, but it is applied on top of a model that already represents ambiguity as part of the label space.

The contribution of this work is **learned deferral as an auditable decision-control interface**. The model does not generate free-form decisions. It emits bounded labels, confidence scores, and structured auxiliary signals, while the release provides the artifacts needed to inspect behavior: calibration, threshold sweeps, multi-seed stability, confusion matrices, and reproducibility materials.

Contributions.

- We formulate decision routing as bounded control over YES, NO, and learned TBD deferral.
- We present a transformer-based multi-head architecture that separates primary decision prediction from structured auxiliary context signals.
- We describe targeted boundary refinement for hard ambiguity cases, including negation, contradiction, lexical-overlap traps, and modal/hedged statements.
- We evaluate the S12B publish candidate on full validation and test splits and report per-class behavior, calibration, threshold-sweep evidence, and stability checks.
- We provide a certifier-facing artifact structure that makes evaluation results, calibration data, threshold sweeps, and reproducibility materials available as a coherent review package.

2 Related Work

EvaluatorDPT is related to selective classification and reject-option models, which abstain from prediction under uncertainty [5, 7, 6]. In many such systems, abstention is applied after prediction using confidence thresholds. EvaluatorDPT instead learns a deferral label directly while still preserving threshold sweeps for operating-point selection. A learned TBD class allows ambiguous examples to shape the representation during training, while thresholding remains a deployment policy rather than the only source of abstention.

The architecture builds on transformer language models such as BERT [4] and multi-task learning [2]. Here, auxiliary outputs are treated as structured context signals available at inference time rather than only as training regularizers.

Neural confidence scores can be miscalibrated [9], so EvaluatorDPT pairs decision outputs with calibration evidence and threshold-sweep artifacts. This is relevant for governance because the operating policy should be inspectable rather than implicit.

The paper is also related to model cards and governance-oriented reporting practices. Such artifacts often document intended use, limitations, and evaluation results after training. EvaluatorDPT treats this material as part of the release surface: the model is accompanied by a structured certification pack containing metrics, calibration data, threshold sweeps, stability checks, and reproducibility commands.

3 System Overview

The model receives a context signal and returns:

- a bounded decision: YES, NO, or TBD;
- decision confidence derived from the primary decision head;
- structured auxiliary signals for values and emotions/sentiments, where enabled by the lineage;
- evidence artifacts that support threshold selection and audit.

The intended deployment pattern is conservative. YES and NO represent bounded action signals. TBD represents a case for human review, a policy engine, additional evidence collection, or another fallback. Because the outputs are bounded, downstream systems can log the decision distribution and threshold policy without parsing generated text.

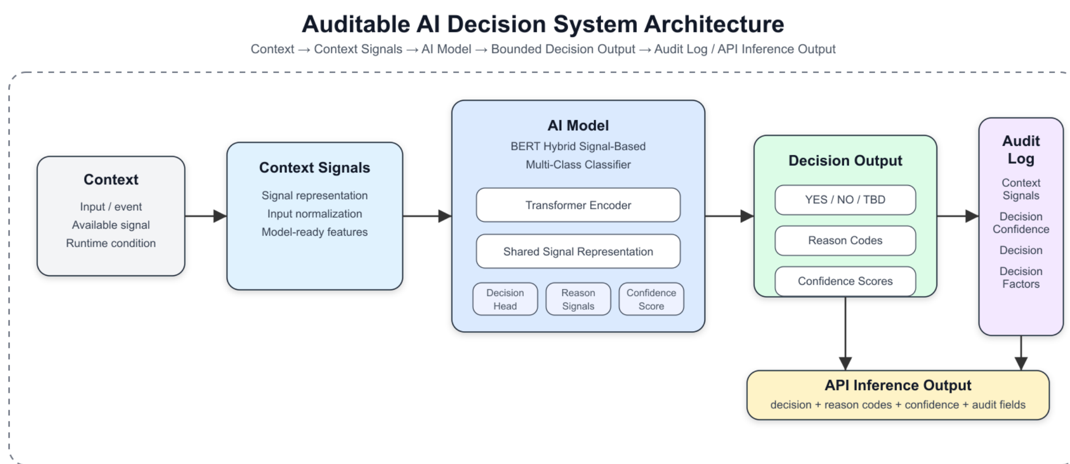

Figure 1: EvaluatorDPT system architecture: tokenized decision event to shared transformer encoder, bounded decision output, confidence, and structured auxiliary channels.

4 Methodology

4.1 Bounded Decision Formulation

Let x be a decision-event input. EvaluatorDPT predicts a primary decision distribution over $\mathcal{Y}_d = \{\text{YES}, \text{NO}, \text{TBD}\}$. The TBD label represents explicit deferral under insufficient semantic certainty. This differs from binary classification followed by confidence filtering because the model learns examples of deferral as part of the supervised task.

Deployment policies may still choose operating thresholds over the predicted probabilities. In this formulation, thresholding is an operating policy layered over a bounded decision model, not the sole mechanism by which uncertainty is represented.

This formulation gives the system two mechanisms for risk control. The first is learned: ambiguous or insufficient contexts can be labeled TBD during training. The second is operational: deployment owners can choose thresholds over the learned distribution to match risk tolerance. A high-assurance setting may route more cases to TBD, while a lower-risk setting may accept a smaller margin for YES or NO. The model separates the representation of uncertainty from the policy decision about how much uncertainty is acceptable.

4.2 Architecture

EvaluatorDPT uses a BERT-base encoder [4] with a maximum sequence length of 128 tokens. The shared encoder feeds task-specific heads:

- a primary three-class decision head;
- a 10-label value head;
- a 28-label emotion/sentiment head.

For S12B, the decision head is the validated output reported in this paper. The emotion head is masked in the DS-L lineage, so emotion metrics are intentionally not claimed for this release candidate. The auxiliary-head architecture remains part of the interface design, but this paper scopes empirical claims to the validated S12B decision behavior and certification evidence.

4.3 Data and Source Scope

The DS-L lineage is a curated decision-event corpus built from public NLP sources and curated decision examples. The corpus is organized around bounded decision labels and auxiliary context signals. The release does not redistribute raw training data. Instead, it provides dataset documentation, processing notes, and artifact pointers sufficient for review of the published model candidate.

The dataset should be interpreted as a research and evaluation corpus for bounded decision behavior, not as a universal production benchmark. The paper evaluates learned deferral and release artifacts under the reported distribution; it does not claim universal generalization to all domains.

4.4 Boundary Refinement

A recurring challenge in bounded decision systems is the boundary between action and deferral. S12B uses DS-L with a targeted boundary pack focused on hard semantic regions such as contradiction, negation, low lexical overlap entailment, high lexical overlap contradiction, long-text stress cases, and hedge/modal ambiguity. The goal is to improve boundary behavior using targeted examples rather than relying only on broad dataset scale.

This design reflects an empirical lesson from the development lineage: additional data is most useful when it targets the failing boundary. The boundary pack focuses on cases where surface overlap can be misleading, where negation changes the decision, where numbers or roles alter entailment, or where a statement is too hedged to support a direct YES or NO. These cases matter for learned deferral because the model must distinguish confident action from insufficient evidence.

5 Evaluation Setup

The current publish candidate is `exp_t90_S12B_boundarypack_ep1_fromS12ep3`. Results are reported for DS-L validation and test splits:

- validation: n = 44,404;

- test: n = 44,597.

The release artifacts include full-split metrics, confusion matrices, calibration data, threshold sweeps, and reproducibility commands under the public certification bundle.

The primary metric is Macro F1 because the decision labels must be evaluated across all three outcomes rather than dominated by the largest class. Accuracy is reported as a secondary metric. Per-class precision, recall, and F1 are reported to expose tradeoffs among action, rejection, and deferral.

6 Results

6.1 Decision Performance

Split	Accuracy	Macro F1
Validation (n=44,404)	0.8224	0.8213
Test (n=44,597)	0.8260	0.8252

The validation and test results are close, which supports the stability of the reported behavior under the fixed DS-L split. The test Macro F1 of 0.8252 should be interpreted together with the per-class table below because the operational meaning of errors differs across YES, NO, and TBD.

6.2 Per-Class Test Performance

Class	Precision	Recall	F1	Support
YES	0.8205	0.8425	0.8314	14,883
NO	0.8598	0.8376	0.8486	15,650
TBD	0.7955	0.7958	0.7956	14,064

NO is the best-performing class in the S12B test result (F1 = 0.8486), indicating that the model learned a useful rejection boundary. TBD remains the hardest class (F1 = 0.7956), which is expected in a defer-aware system because the deferral region is defined by ambiguity rather than by a single semantic pattern. For this reason, the release includes threshold sweeps and calibration evidence rather than relying only on a single argmax policy.

6.3 Validation Confusion Matrix

	Pred YES	Pred NO	Pred TBD
True YES	12,496	999	1,467
True NO	1,035	13,066	1,433
True TBD	1,793	1,160	10,955

The validation confusion matrix shows that the largest residual issue is confusion between true TBD and direct action labels, especially TBD predicted as YES. This is the main operational boundary for future improvement: reducing premature action while preserving useful YES/NO coverage.

7 Audit and Certification Evidence

EvaluatorDPT is packaged with audit artifacts in addition to evaluation scores. For S12B, the public certification pack is located at:

```
certification/runs/S12B_20260526/
```

The pack includes:

- evaluation and calibration reports;
- calibration data with validation ECE = 0.0338;
- threshold-sweep artifacts for TBD-fallback operating policies;
- multi-seed stability evidence, with validation seeds 42, 0, and 7 producing identical Macro F1 (std = 0.0);
- run metrics, confusion matrix, diagnostics, and reproducibility commands.

This certification structure is part of the system contribution. It exposes the evidence needed to inspect operating behavior rather than only the final checkpoint.

7.1 Why Certification Evidence Matters

For decision-control models, reproducibility and auditability are not secondary documentation tasks. They determine whether a model can be reviewed before deployment. A headline score does not show whether the confidence distribution is calibrated, whether thresholds were selected after inspecting tradeoffs, whether results are stable across seeds, or whether the exact evaluated artifact can be located.

EvaluatorDPT packages evaluation as a set of review artifacts. The certification pack is designed to answer reviewer questions: Which checkpoint was evaluated? Which data split was used? What were the per-class outcomes? What threshold sweeps were run? How stable are the results? Where are the reproducibility commands? This makes the release closer to an operational model handoff than a standalone benchmark table.

8 Discussion

The results show that bounded decision control can be evaluated as a decision interface rather than as ordinary classification alone. S12B achieves balanced performance across YES, NO, and TBD, with NO F1 = 0.8486 and TBD F1 = 0.7956 on the full test split. The remaining difficulty is the boundary between actionable decisions and deferral, especially ambiguous cases where a human reviewer or downstream policy may prefer conservative routing.

Calibration, threshold sweeps, and multi-seed stability make the model's operating behavior inspectable. Downstream systems can select operating thresholds and document tradeoffs without retraining the core model.

The auxiliary architecture also matters, but empirical claims should remain scoped. In S12B, decision behavior is validated; emotion metrics are not reported because the emotion head is masked. Future releases can separately validate auxiliary value and emotion behavior as independent outputs.

The result is not an isolated leaderboard metric. It is a learned-deferral decision model released with artifacts that describe classwise behavior, calibration, threshold tradeoffs, stability, and artifact identity. These materials are necessary when a model is used as a decision interface rather than a passive classifier.

9 Limitations

Results are specific to DS-L, the S12B checkpoint, and the evaluation setup reported here. Additional domains require separate validation. Inputs are truncated to 128 tokens, so long-context use cases require chunking or preprocessing. The emotion head is masked in this lineage, so this paper does not claim validated emotion-head performance for S12B. The dataset composition includes public NLP sources and curated decision examples, but raw training data is not redistributed in this repository. Users should consult original dataset licenses before reusing source datasets.

The paper also does not claim that TBD is always the correct action under uncertainty. TBD is a model output and an operating-policy option, not a substitute for domain governance. Deployment owners must decide how deferred cases are routed and what thresholds are acceptable for their risk environment. Similarly, the reported calibration value is tied to the validation split and should be remeasured after domain transfer, threshold changes, or additional fine-tuning.

Finally, the certification pack should be understood as a review aid, not as a guarantee of universal safety. It makes the evidence inspectable, but each deployment still requires application-specific validation and monitoring.

10 Conclusion

This paper presented EvaluatorDPT, a bounded decision-control model with learned deferral. The contribution is a decision interface: YES, NO, or TBD, paired with confidence, threshold sweeps, calibration evidence, stability checks, and a certifier-facing artifact pack. On the full DS-L test split, S12B achieves Accuracy = 0.8260 and Macro F1 = 0.8252, with per-class F1 scores of 0.8314 (YES), 0.8486 (NO), and 0.7956 (TBD). These results support learned deferral and release-level audit artifacts as practical components for AI decision routing under ambiguity.

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